

# Product Requirements Document

## Badminton Motion Intelligence Platform

**Working title:** Badminton Motion Lab

**Document version:** 0.1

**Status:** Draft for implementation planning

**Product owner:** Rahmad J. E. N. S. S.

**Product type:** Personal hobby project with research-grade foundations

**Primary platform:** Responsive web application and local analysis worker

**Initial operating budget:** Rp0 per month

**Target users:** Individual badminton players, developer-owner, and later badminton coaches

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## 1. Executive Summary

Badminton Motion Intelligence Platform adalah aplikasi yang menganalisis video latihan atau pertandingan badminton untuk menghasilkan pengukuran teknik, footwork, timing, konsistensi, dan pola permainan yang dapat ditelusuri kembali ke frame video.

Produk tidak diposisikan sebagai AI yang langsung menentukan apakah teknik pemain “benar” atau “salah”. Produk diposisikan sebagai alat pengukuran dan evaluasi berbasis bukti yang:

- Menunjukkan apa yang terdeteksi.
- Menjelaskan bagaimana metrik dihitung.
- Menampilkan frame sumber.
- Menyatakan tingkat keyakinan.
- Mengungkap keterbatasan analisis.
- Memungkinkan koreksi manual.
- Membandingkan perkembangan pemain antar-sesi.

Visi produk mencakup analisis single-camera, multi-camera, stroke technique, footwork, rally, pertandingan penuh, tactical intelligence, athlete progress, dan coach workspace.

Implementasi dilakukan secara bertahap, tetapi seluruh fondasi data, pipeline, versioning, annotation, dan validation harus disiapkan sejak awal.

Untuk menjaga biaya awal mendekati nol, produk menggunakan pendekatan:

**Local-first analysis, cloud-light synchronization, and optional pay-per-use compute.**

Video asli diproses pada komputer pengguna atau developer. Cloud hanya digunakan untuk web interface, autentikasi, metadata, hasil analisis, dan artifact kecil. Cloud GPU tidak menjadi ketergantungan awal.

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## 2. Problem Statement

### 2.1 Masalah Pengguna

Pemain badminton rekreasional biasanya mengevaluasi gerakan melalui:

- Perasaan pribadi.
- Video slow motion.
- Komentar teman.
- Penilaian pelatih yang tidak selalu tersedia.
- Perbandingan visual yang subjektif.

Video biasa dapat menunjukkan bahwa suatu gerakan “terlihat berbeda”, tetapi tidak memberikan pengukuran yang konsisten tentang:

- Sudut tubuh.
- Timing gerakan.
- Posisi contact.
- Footwork path.
- Landing balance.
- Recovery.
- Konsistensi antar-repetisi.
- Perubahan performa antar-sesi.

Sebagian aplikasi computer vision hanya menampilkan skeleton overlay atau skor tanpa menjelaskan bagaimana skor diperoleh. Hal ini menciptakan kesan presisi tanpa validitas yang cukup.

### 2.2 Masalah Teknis

Analisis badminton memiliki tantangan khusus:

- Gerakan sangat cepat.
- Tangan, raket, dan shuttlecock sering mengalami motion blur.
- Anggota tubuh saling menutupi.
- Posisi tubuh dapat ekstrem.
- Video pengguna memiliki sudut dan kualitas yang berbeda.
- Single-camera memiliki depth ambiguity.
- Tidak semua metrik dapat dihitung dari semua video.
- Definisi “teknik ideal” dapat berbeda menurut pelatih dan konteks.

Karena itu, produk membutuhkan lebih dari pose estimation. Produk harus memiliki:

- Capture-quality assessment.
- Court calibration.
- Body-pose tracking.
- Racket tracking.

- Shuttle tracking.
  - Temporal event detection.
  - Deterministic metric engine.
  - Assessment rubric.
  - Confidence propagation.
  - Validation framework.
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## 3. Product Vision

Membangun platform personal yang mampu mengubah video badminton menjadi data gerakan yang terstruktur, terukur, dapat dibandingkan, dan semakin akurat melalui koreksi serta validasi.

### 3.1 Long-Term Product Statement

A personal badminton motion and performance laboratory that transforms training and match videos into measurable, evidence-backed, and progressively validated insights.

### 3.2 North-Star Experience

Seorang pemain merekam latihan menggunakan smartphone, membuka aplikasi, memilih video, dan memperoleh:

1. Pemeriksaan kualitas video.
2. Deteksi pemain dan court.
3. Skeleton serta lintasan tubuh.
4. Segmentasi stroke atau rally.
5. Deteksi fase gerakan.
6. Pengukuran teknik dan footwork.
7. Evidence frame untuk setiap temuan.
8. Confidence serta limitation.
9. Perbandingan dengan sesi sebelumnya.
10. Prioritas latihan berikutnya.

Pengguna dapat mengoreksi event atau landmark yang salah. Koreksi tersebut meningkatkan kualitas histori pengguna dan dapat dimasukkan ke dataset pengembangan setelah mendapat persetujuan.

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## 4. Goals

### 4.1 Product Goals

1. Menghasilkan analisis gerakan badminton yang transparan dan dapat ditelusuri.
2. Memungkinkan pengguna melihat perkembangan secara kuantitatif.

3. Memisahkan hasil pengukuran dari interpretasi teknik.
4. Menyediakan pengalaman yang tetap berguna tanpa cloud GPU.
5. Mengembangkan dataset badminton personal secara bertahap.
6. Mendukung peningkatan dari single-camera menuju multi-camera.
7. Menjadikan koreksi manusia sebagai bagian inti sistem.
8. Memastikan setiap klaim presisi memiliki benchmark yang sesuai.
9. Menjaga biaya infrastruktur awal sebesar Rp0 per bulan.
10. Menghindari ketergantungan pada satu model AI atau penyedia cloud.

## 4.2 Engineering Goals

- Pipeline dapat dijalankan secara lokal.
- Setiap analysis run dapat direproduksi.
- Model dapat diganti melalui adapter.
- Raw data tidak ditimpa.
- Metric definition memiliki versi.
- Analysis dapat dijalankan ulang sebagian.
- Semua output memiliki provenance.
- Aplikasi tetap dapat digunakan ketika worker sedang offline.
- Cloud storage bukan syarat untuk menyimpan video asli.
- Perpindahan ke cloud worker tidak mengubah domain model.

## 4.3 Research Goals

- Membentuk benchmark video badminton milik sendiri.
- Mengukur pose error, event error, dan repeatability.
- Membandingkan beberapa pose model.
- Menilai pengaruh camera angle dan frame rate.
- Mengembangkan badminton-specific event detector.
- Memvalidasi selected metrics dengan multi-camera atau expert annotation.

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## 5. Non-Goals

Produk tidak bertujuan untuk:

- Memberikan diagnosis medis.
- Menentukan risiko cedera secara klinis.
- Menggantikan fisioterapis atau tenaga kesehatan.
- Mengklaim force, joint moment, muscle load, atau power aktual hanya dari video biasa.
- Mengklaim kecepatan shuttle aktual tanpa kalibrasi yang memadai.
- Memberikan skor “teknik sempurna” yang berlaku universal.
- Menyembunyikan uncertainty.
- Mengunggah video pengguna ke dataset training secara otomatis.
- Membangun social network pada fase awal.
- Membangun sistem pembayaran atau subscription.

- Menjadi live line-calling system.
  - Menjadi pengganti pelatih profesional.
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## **6. Product Principles**

### **6.1 Evidence Before Advice**

Setiap feedback harus berasal dari observation dan metric yang dapat dilihat kembali pada video.

### **6.2 Measurement Before Judgment**

Sistem terlebih dahulu mengukur. Penilaian dilakukan melalui rubric terpisah.

### **6.3 Local First**

Video asli dan proses komputasi berat berjalan secara lokal secara default.

### **6.4 Honest Uncertainty**

Sistem diperbolehkan mengatakan bahwa sebuah metrik tidak dapat dihitung.

### **6.5 Progressive Precision**

Semakin baik capture mode, semakin banyak metrik dan semakin kuat klaim akurasi.

### **6.6 Human-Correctable**

Semua event penting harus dapat dikoreksi pengguna atau reviewer.

### **6.7 Version Everything**

Model, metric, rubric, pipeline, annotation, dan hasil analisis harus memiliki versi.

### **6.8 Modular Intelligence**

Pose, racket, shuttle, temporal detection, metrics, dan assessment merupakan modul yang terpisah.

### **6.9 No Fabricated Metrics**

LLM atau UI tidak boleh menghasilkan angka yang tidak tersedia dari pipeline.

## 6.10 Cost Is an Architectural Constraint

Fitur yang membutuhkan biaya rutin tidak boleh menjadi dependency bagi pengalaman inti pada fase awal.

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# 7. Target Users

## 7.1 Primary Persona — Personal Badminton Player

### Profile

- Bermain badminton sebagai hobi.
- Merekam drill atau permainan menggunakan smartphone.
- Ingin mengetahui perkembangan gerakan.
- Tidak memiliki akses rutin ke sport-science laboratory.
- Bersedia mengikuti capture guide sederhana.

### Primary needs

- Melihat gerakan secara slow motion.
- Memahami bagian gerakan yang tidak konsisten.
- Membandingkan sesi.
- Mendapat pengukuran yang mudah dipahami.
- Mengetahui apakah hasil analisis cukup terpercaya.

## 7.2 Secondary Persona — Developer/Researcher

### Profile

- Mengembangkan dan menguji pipeline.
- Menjalankan model pada komputer sendiri.
- Membuat annotation.
- Membandingkan model.
- Meninjau error per frame.

### Primary needs

- Reproducible analysis.
- Model versioning.
- Dataset management.
- Benchmarking.
- Debug overlays.
- Export data.

## 7.3 Future Persona — Badminton Coach

### Profile

- Meninjau teknik beberapa pemain.
- Memiliki rubric atau pendekatan coaching sendiri.
- Membutuhkan evidence frame dan comparison.

### Primary needs

- Mengoreksi event.
  - Memberi komentar.
  - Menentukan reference range.
  - Membandingkan athlete sessions.
  - Menyetujui atau menolak finding.
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## 8. Jobs to Be Done

### JTBD-01 — Understand a Movement

Ketika saya merekam satu atau beberapa pukulan, saya ingin melihat fase dan gerakan tubuh secara terukur agar saya memahami apa yang sebenarnya terjadi.

### JTBD-02 — Compare Sessions

Ketika saya berlatih kembali, saya ingin membandingkan hasil dengan sesi sebelumnya agar saya mengetahui apakah perubahan saya nyata atau hanya terasa berbeda.

### JTBD-03 — Find Inconsistency

Ketika saya melakukan beberapa repetisi, saya ingin mengetahui bagian yang paling tidak konsisten agar latihan saya lebih terarah.

### JTBD-04 — Validate a Finding

Ketika aplikasi memberikan feedback, saya ingin melihat frame, nilai, confidence, dan reference yang digunakan agar saya dapat menilai apakah feedback tersebut masuk akal.

### JTBD-05 — Correct the System

Ketika sistem salah menentukan stroke atau contact frame, saya ingin memperbaikinya tanpa kehilangan hasil asli.

## JTBD-06 — Improve the Model

Ketika saya telah mengoreksi beberapa video, saya ingin menggunakan data tersebut untuk menguji dan meningkatkan pipeline.

## JTBD-07 — Analyze Without Cloud Cost

Ketika saya menjalankan analisis, saya ingin menggunakan komputer sendiri agar tidak membayar cloud GPU untuk setiap video.

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# 9. Product Scope

## 9.1 Complete Product Scope

Produk akhir mencakup enam domain utama:

1. **Capture Studio**
2. **Technique Lab**
3. **Footwork Lab**
4. **Match Intelligence**
5. **Progress and Comparison**
6. **Coach and Research Workspace**

## 9.2 Implementation Scope by Capability

| Capability | Foundation                      | Advanced                                    |
|------------|---------------------------------|---------------------------------------------|
| Capture    | Upload, metadata, quality guide | Live positioning and capture validation     |
| Pose       | 2D body pose                    | Whole-body and badminton fine-tuning        |
| Court      | Manual calibration              | Automatic line detection                    |
| Racket     | Manual marker                   | Automatic head and shaft tracking           |
| Shuttle    | Manual contact marker           | Automatic trajectory tracking               |
| Stroke     | Manual segmentation             | Automatic hierarchical classification       |
| Technique  | Basic deterministic metrics     | Full phase and sequence analysis            |
| Footwork   | Body path and basic step events | Detailed foot type and efficiency           |
| Match      | Manual rally structure          | Full automatic match intelligence           |
| Precision  | Single-camera                   | Multi-camera reconstruction                 |
| Feedback   | Structured rule-based findings  | Personalized and coach-approved explanation |



| Capability | Foundation               | Advanced                          |
|------------|--------------------------|-----------------------------------|
| Validation | Internal annotated clips | Multi-camera and expert benchmark |
| Compute    | Local worker             | Optional cloud workers            |

## 10. Capture Modes

### 10.1 Quick Analysis

#### Input

- One smartphone video.
- Informal camera placement.
- No detailed calibration.

#### Supported outputs

- Body pose.
- Basic timing.
- Relative movement.
- Basic consistency.
- Visual comparison.

#### Restrictions

- No strong 3D claim.
- Limited contact and racket analysis.
- Limited absolute distance.

### 10.2 Guided Single-Camera

#### Input

- Static camera.
- Capture guide followed.
- Court lines visible.
- Player fully visible.
- Camera position recorded.

#### Supported outputs

- Court-plane position.
- Footwork path.
- Better joint-angle interpretation.

- Better contact estimate.
- Session comparison.

## 10.3 Multi-Camera

### Input

- Two or more synchronized videos.
- Camera calibration.
- Shared court reference.

### Supported outputs

- Triangulated 3D keypoints.
- Stronger kinematic measurement.
- Racket orientation.
- Contact point in 3D.
- Better body sequencing.

## 10.4 Validation Mode

### Input

- Multi-camera and optional external sensor.
- Expert annotation.
- Controlled capture protocol.

### Purpose

- Model validation.
- Metric validation.
- Dataset creation.
- Error measurement.

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# 11. Core User Flow

## 11.1 Create Session

1. User selects **New Session**.
2. User chooses:
3. Drill.
4. Free play.

5. Match.
6. Validation session.
7. User enters:
8. Session name.
9. Date.
10. Dominant hand.
11. Capture mode.
12. Camera angle.
13. Intended analysis.
14. Application creates a session record.

## 11.2 Add Video

User can choose one of three methods:

### Local Analysis

- Select file through local worker.
- Video remains on the device.
- Cloud receives metadata and analysis output only.

### Cloud-Synced Small Clip

- Upload a short, compressed clip.
- Used only when it fits within configured quota.

### External Object Storage

- Upload to optional S3-compatible storage.
- Enabled only after explicit setup.

## 11.3 Quality Check

System analyzes:

- Resolution.
- Frame rate.
- Duration.
- Orientation.
- Body visibility.
- Player size.
- Camera stability.
- Blur.
- Lighting.
- Court visibility.

- Number of visible players.

System returns:

- Supported.
- Supported with limitations.
- Low confidence.
- Unsupported.

## 11.4 Calibration

Depending on mode:

- User identifies court corners.
- System detects court lines.
- User corrects detected lines.
- Camera angle is recorded.
- Player height may be entered as optional reference.

## 11.5 Run Analysis

User chooses:

- Fast preview.
- Standard local analysis.
- Detailed local analysis.
- Custom research pipeline.

The application creates an immutable `analysis_run`.

## 11.6 Review Results

User sees:

- Video player.
- Skeleton overlay.
- Event timeline.
- Joint metrics.
- Footwork path.
- Confidence.
- Quality limitations.
- Findings.
- Comparison.

## 11.7 Correct Results

User can correct:

- Player identity.
- Court corners.
- Stroke range.
- Contact frame.
- Stroke label.
- Landmark.
- Foot contact.
- Excluded frame.

## 11.8 Save and Compare

Corrected results are saved as a new annotation revision. Original model output remains unchanged.

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# 12. Functional Requirements

Priority definitions:

- **P0:** Required foundation.
  - **P1:** Required for a useful personal product.
  - **P2:** Advanced product capability.
  - **P3:** Research or future expansion.
- 

## 12.1 Identity and Profile

### IDN-001 — User Authentication

**Priority:** P0

The system shall allow a user to sign in and access only their own data.

#### Acceptance criteria

- User can create a session.
- User cannot read another user's private session.
- Row-level authorization is enabled.
- Local-only development mode may use a seeded developer account.

## **IDN-002 — Athlete Profile**

**Priority:** P0

The system shall store:

- Display name.
- Dominant hand.
- Optional height.
- Playing level.
- Optional coach notes.
- Consent settings.

## **IDN-003 — Multiple Athlete Support**

**Priority:** P1

A user may create more than one athlete profile without changing the core database structure.

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# **12.2 Session Management**

## **SES-001 — Create and Edit Session**

**Priority:** P0

A user shall create a session with session type, date, capture mode, and objective.

## **SES-002 — Session Status**

**Priority:** P0

Supported statuses:

- Draft.
- Ready.
- Processing.
- Needs review.
- Complete.
- Failed.
- Archived.

## **SES-003 — Session Timeline**

**Priority:** P1

A session shall show all videos, analysis runs, corrections, and reports in chronological order.

## **SES-004 — Session Comparison**

**Priority:** P1

A user shall compare two or more compatible sessions.

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## **12.3 Media Management**

### **MED-001 — Preserve Original Media**

**Priority:** P0

Original files shall never be overwritten by normalization or rendering.

### **MED-002 — Local Media Reference**

**Priority:** P0

The platform shall support media that remains on the user's local machine.

A local reference shall store:

- File fingerprint.
- Filename.
- File size.
- Duration.
- Resolution.
- Frame rate.
- Last known local path only when explicitly permitted.

### **MED-003 — Derived Assets**

**Priority:** P0

Derived assets may include:

- Proxy video.
- Thumbnail.
- Frame image.
- Normalized video.
- Overlay video.
- Analysis package.
- JSON result.
- CSV export.

## **MED-004 — Cloud Media Is Optional**

**Priority:** P0

The application shall remain functional when no raw video is stored in cloud storage.

## **MED-005 — Retention Policy**

**Priority:** P1

User can choose:

- Keep original locally.
  - Delete local derived frames after processing.
  - Keep only metrics.
  - Delete cloud artifacts.
  - Delete entire session.
- 

# **12.4 Local Analysis Worker**

## **WRK-001 — Local Worker**

**Priority:** P0

A Python worker shall process videos on the developer's or user's machine.

## **WRK-002 — Worker Health**

**Priority:** P0

The web application shall display:

- Connected.
- Busy.
- Offline.
- Version incompatible.

## **WRK-003 — Job Lease**

**Priority:** P0

A worker shall claim a job for a limited period so the same job is not processed twice unintentionally.



## **WRK-004 — Idempotent Processing**

**Priority:** P0

A pipeline step shall not duplicate output when safely retried.

## **WRK-005 — Partial Reprocessing**

**Priority:** P1

A user shall rerun only:

- Pose.
- Court detection.
- Event detection.
- Metric calculation.
- Rendering.

without rerunning unrelated steps.

## **WRK-006 — Hardware Profile**

**Priority:** P1

The worker shall record:

- Operating system.
- CPU.
- Available RAM.
- GPU type when available.
- Processing backend.
- Model execution time.

## **WRK-007 — Offline Analysis Package**

**Priority:** P1

The worker shall be able to export a portable analysis package that can later be imported into the web application.

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## **12.5 Quality Assessment**

### **QLT-001 — Media Technical Validation**

**Priority:** P0

The system shall detect:

- Unsupported codec.
- Invalid duration.
- Corrupted stream.
- Variable frame rate.
- Rotation metadata.
- Very low resolution.

## **QLT-002 — Visual Quality Check**

**Priority:** P1

The system shall estimate:

- Motion blur.
- Lighting.
- Body visibility.
- Camera movement.
- Player scale.
- Court visibility.

## **QLT-003 — Analysis Eligibility Matrix**

**Priority:** P0

The quality report shall state which analyses are:

- Available.
- Available with reduced confidence.
- Unavailable.

## **QLT-004 — Quality Flags**

**Priority:** P0

Every affected metric shall inherit relevant quality flags.

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# **12.6 Court and Coordinate System**

## **CRT-001 — Manual Court Calibration**

**Priority:** P0

A user shall identify court reference points manually.

## **CRT-002 — Automatic Court Detection**

**Priority:** P1

The system shall propose court lines and corners for correction.

## **CRT-003 — Coordinate Transform**

**Priority:** P0

The system shall support conversion between:

- Image pixels.
- Normalized image coordinates.
- Court coordinates.
- Camera coordinates.
- 3D world coordinates where applicable.

## **CRT-004 — Coordinate Metadata**

**Priority:** P0

No coordinate may be stored without:

- Coordinate-system identifier.
  - Unit.
  - Source.
  - Transform version.
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# **12.7 Pose Estimation**

## **POS-001 — Pose Model Adapter**

**Priority:** P0

Pose models shall implement a common output contract.

## **POS-002 — Browser Pose Preview**

**Priority:** P1

A lightweight browser model may provide:

- Capture framing.
- Whole-body visibility check.

- Fast skeleton preview.

MediaPipe Pose Landmarker supports browser and JavaScript processing for image and video, produces 33 landmarks, and provides image as well as estimated 3D world coordinates. Its video calls are synchronous, so browser implementation should execute inference through a Web Worker to avoid blocking the interface.

### **POS-003 — Local Detailed Pose**

**Priority:** P0

The local worker shall support a server-grade or desktop-grade pose engine.

MMPose exposes an inference interface for images and videos and provides multiple pre-trained model families, making it suitable behind a model adapter rather than tightly coupling the application to one checkpoint.

### **POS-004 — Landmark Confidence**

**Priority:** P0

Every landmark observation shall store:

- X.
- Y.
- Optional Z.
- Visibility.
- Confidence.
- Frame timestamp.
- Source model.

### **POS-005 — Raw and Smoothed Pose**

**Priority:** P0

Raw and temporally smoothed keypoints shall remain separate.

### **POS-006 — Pose Correction**

**Priority:** P1

A reviewer shall be able to reposition selected landmarks on selected frames.

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## 12.8 Racket and Shuttle

### RKT-001 — Manual Racket Marker

**Priority:** P1

Before automatic tracking is reliable, a user shall be able to mark:

- Racket head.
- Handle.
- Shaft direction.

### RKT-002 — Automatic Racket Tracking

**Priority:** P2

The system shall track racket head and shaft with confidence per frame.

### SHT-001 — Manual Contact Marker

**Priority:** P0

A user shall be able to define the estimated shuttle-contact frame.

### SHT-002 — Shuttle Tracking Adapter

**Priority:** P2

The shuttle tracker shall implement a replaceable model interface.

TrackNetV3 is a potential baseline because it separates trajectory prediction and trajectory rectification and is distributed under an MIT license. It shall be evaluated on personal smartphone footage before being adopted as a production default.

### SHT-003 — Shuttle Track Confidence

**Priority:** P2

Interpolated shuttle points shall be explicitly distinguishable from observed points.

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## 12.9 Temporal and Badminton Events

### EVT-001 — Manual Stroke Segmentation

**Priority:** P0

A user shall define stroke start, contact, and end.

### **EVT-002 — Automatic Stroke Proposal**

**Priority:** P1

The system shall propose stroke boundaries for user confirmation.

### **EVT-003 — Movement Phases**

**Priority:** P1

Supported phases include:

- Ready.
- Preparation.
- Backswing.
- Acceleration.
- Contact.
- Follow-through.
- Landing.
- Recovery.

### **EVT-004 — Rally Structure**

**Priority:** P2

The system shall represent:

- Match.
- Game.
- Rally.
- Stroke.
- Phase.
- Frame observation.

### **EVT-005 — Hierarchical Stroke Taxonomy**

**Priority:** P0

Stroke shall be represented using attributes rather than one rigid class.

Minimum fields:

- Stroke family.
- Subtype.
- Forehand/backhand.

- Contact height.
  - Court zone.
  - Straight/cross-court.
  - Jump/non-jump.
  - Outcome.
- 

## 12.10 Metric Engine

### MET-001 — Deterministic Metrics

**Priority:** P0

Metric values shall be computed by versioned functions rather than generated by an LLM.

### MET-002 — Metric Definition

**Priority:** P0

Each metric definition shall include:

- Name.
- Description.
- Formula or algorithm reference.
- Unit.
- Required observations.
- Supported capture modes.
- Metric version.
- Known limitations.

### MET-003 — Evidence Frames

**Priority:** P0

Each metric observation shall reference supporting frames or time ranges.

### MET-004 — Confidence and Uncertainty

**Priority:** P0

Each observation shall store:

- Confidence.
- Optional uncertainty interval.
- Quality flags.
- Missing input indicators.

## **MET-005 — Technique Metrics**

**Priority:** P1

Initial technique metric catalogue:

- Elbow angle.
- Shoulder elevation.
- Trunk inclination.
- Shoulder-hip separation.
- Knee flexion.
- Reach distance in image plane.
- Contact position relative to body.
- Preparation duration.
- Swing duration.
- Follow-through range.
- Recovery duration.

## **MET-006 — Footwork Metrics**

**Priority:** P1

Initial footwork metric catalogue:

- Player path.
- Travel distance on court plane.
- Base-position deviation.
- Time to first movement.
- Maximum lunge position.
- Landing position.
- Recovery path.
- Recovery duration.
- Left-right asymmetry proxy.

## **MET-007 — Consistency Metrics**

**Priority:** P1

For repeated strokes, system shall calculate:

- Mean.
- Median.
- Range.
- Standard deviation.
- Coefficient of variation where meaningful.
- Contact-point dispersion.
- Event-timing dispersion.



## **MET-008 — Kinetic Proxy Labeling**

**Priority:** P0

Video-derived estimates shall not be labeled as actual force or power unless validated.

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## **12.11 Assessment Engine**

### **ASM-001 — Separation from Measurement**

**Priority:** P0

Assessment shall consume metrics rather than raw video.

### **ASM-002 — Reference Types**

**Priority:** P1

The system shall support:

- Personal baseline.
- Coach-defined reference.
- Research reference.
- Population reference in a later phase.

### **ASM-003 — Transparent Finding**

**Priority:** P0

Every finding shall display:

- Metric.
- Observed value.
- Reference value or comparison.
- Direction of difference.
- Confidence.
- Evidence.
- Limitation.
- Rubric version.

### **ASM-004 — No Universal Technique Score**

**Priority:** P0

The product shall not present one overall technique score until its construction and validation are documented.

## **ASM-005 — Structured Recommendation**

**Priority:** P1

Recommendations shall map to one or more validated findings.

---

## **12.12 Video Review Interface**

### **VWR-001 — Frame-Accurate Playback**

**Priority:** P0

The viewer shall support:

- Play.
- Pause.
- Frame step.
- Playback speed.
- Timecode.
- Event jump.

### **VWR-002 — Overlay Layers**

**Priority:** P0

User can toggle:

- Body skeleton.
- Joint labels.
- Joint angles.
- Court.
- Player path.
- Racket.
- Shuttle.
- Events.
- Confidence indicators.

### **VWR-003 — Event Timeline**

**Priority:** P0

Stroke phases and key events shall appear on an editable timeline.

### **VWR-004 — Side-by-Side Comparison**

**Priority:** P1

Two videos shall be synchronized by:

- Contact.
- Stroke start.
- Landing.
- Custom event.

### **VWR-005 — Overlay Comparison**

**Priority:** P2

Two normalized movement sequences may be overlaid with clear source distinction.

---

## **12.13 Annotation and Review**

### **ANN-001 — Non-Destructive Annotation**

**Priority:** P0

Human correction shall not overwrite original model output.

### **ANN-002 — Annotation Revision**

**Priority:** P0

Every correction shall store:

- Reviewer.
- Timestamp.
- Previous value.
- New value.
- Reason.
- Revision number.

### **ANN-003 — Review Status**

**Priority:** P1

Annotation levels:

- Unreviewed.
- User corrected.
- Reviewed.
- Expert reviewed.
- Reference validated.

## **ANN-004 — Export Annotation**

**Priority:** P1

Annotations shall be exportable to a documented JSON format.

## **ANN-005 — Training Consent**

**Priority:** P0

Correction data shall not enter a training dataset unless the user explicitly opts in.

---

# **12.14 Progress and Comparison**

## **PRG-001 — Session History**

**Priority:** P1

User shall view metrics over time.

## **PRG-002 — Comparable Analysis Filter**

**Priority:** P0

The system shall warn when two results are not directly comparable because of:

- Different capture modes.
- Different camera angles.
- Different metric versions.
- Different analysis models.
- Low-quality input.

## **PRG-003 — Personal Baseline**

**Priority:** P1

The system shall calculate a baseline from selected valid sessions.

## **PRG-004 — Best Repetition**

**Priority:** P1

User shall mark or calculate a best repetition using selected criteria.

## **PRG-005 — Trend Evidence**

**Priority:** P1

A trend shall show the number of valid observations and uncertainty, not only a line chart.

---

## **12.15 Reports and Export**

### **RPT-001 — Session Report**

**Priority:** P1

A report shall include:

- Session metadata.
- Capture quality.
- Main metrics.
- Findings.
- Evidence frames.
- Limitations.
- Analysis version.

### **RPT-002 — Data Export**

**Priority:** P0

User shall export:

- JSON.
- CSV.
- Analysis manifest.
- Optional overlay video.

### **RPT-003 — Reproducibility Manifest**

**Priority:** P0

Export shall include model and pipeline versions required to reproduce the output.

---

# 13. LLM Requirements

## 13.1 Allowed Uses

An LLM may:

- Summarize structured findings.
- Explain a metric in plain language.
- Compare sessions based on provided values.
- Convert coach notes into a structured draft.
- Generate a training-summary narrative.
- Prioritize existing findings.

## 13.2 Prohibited Uses

An LLM shall not:

- Infer joint coordinates directly from the uploaded video.
- Invent a metric.
- Calculate the official value when deterministic code is available.
- Hide missing data.
- Diagnose injury.
- Override a low-confidence warning.
- Claim that a movement is universally correct.
- make unsupported claims about force, muscle activation, or injury risk.

## 13.3 Grounding Contract

The LLM shall receive only structured records such as:

```
{
  "metric_key": "recovery_duration",
  "value": 1.18,
  "unit": "seconds",
  "comparison_value": 0.93,
  "comparison_type": "personal_best",
  "confidence": 0.84,
  "evidence": {
    "start_frame": 822,
    "end_frame": 867
  },
  "quality_flags": [
    "left_foot_partial_occlusion"
  ]
}
```

```
]
}
```

All numeric statements in generated text must exist in the supplied structure.

LLM integration is optional and disabled by default to preserve the Rp0 operating budget.

---

## 14. Information Architecture

### 14.1 Main Navigation

- Dashboard.
- Sessions.
- Technique Lab.
- Footwork Lab.
- Match Analysis.
- Comparisons.
- Progress.
- Capture Guide.
- Research.
- Settings.

Advanced modules may remain hidden until implemented.

### 14.2 Session Detail Tabs

- Overview.
- Media.
- Quality.
- Calibration.
- Analysis.
- Metrics.
- Findings.
- Comparison.
- Annotations.
- Runs.
- Export.

### 14.3 Research Workspace

- Dataset browser.
- Annotation queue.
- Model registry.
- Benchmark runs.

- Error cases.
  - Validation reports.
- 

## 15. Data Model Requirements

Core entity groups:

### 15.1 Identity

- users
- athletes
- coaches
- athlete\_coach\_links
- consents

### 15.2 Sessions and Media

- sessions
- captures
- media\_assets
- capture\_profiles
- camera\_calibrations
- synchronization\_groups

### 15.3 Processing

- analysis\_jobs
- analysis\_runs
- pipeline\_steps
- model\_versions
- processing\_artifacts
- quality\_checks

### 15.4 Perception

- detected\_entities
- entity\_tracks
- pose\_sequences
- landmark\_observations
- racket\_tracks
- shuttle\_tracks
- court\_detections
- coordinate\_transforms



## 15.5 Badminton Domain

- matches
- games
- rallies
- strokes
- movement\_events
- contact\_events
- foot\_contacts
- shot\_outcomes

## 15.6 Analytics

- metric\_definitions
- metric\_observations
- assessment\_rubrics
- assessments
- findings
- comparison\_sets

## 15.7 Annotation and Validation

- annotation\_tasks
- annotations
- annotation\_revisions
- review\_decisions
- benchmark\_datasets
- benchmark\_cases
- validation\_results

Detailed ERD shall be produced as a separate technical specification.

---

# 16. Analysis Pipeline Contract

Every pipeline step shall accept a versioned input manifest and produce a versioned artifact manifest.

## 16.1 Pipeline Stages

1. Media inspection.
2. Media normalization.
3. Quality assessment.
4. Player detection.
5. Court calibration.
6. Pose estimation.

7. Pose smoothing.
8. Racket detection.
9. Shuttle tracking.
10. Temporal segmentation.
11. Event detection.
12. Coordinate transformation.
13. Metric calculation.
14. Assessment.
15. Rendering.
16. Report generation.

## 16.2 Required Step Metadata

Each step records:

- Step key.
- Step version.
- Status.
- Start and finish time.
- Input artifact hashes.
- Output artifact hashes.
- Model version.
- Runtime environment.
- Errors and warnings.
- Processing duration.

## 16.3 Statuses

- Queued.
- Claimed.
- Running.
- Complete.
- Complete with warnings.
- Failed.
- Cancelled.
- Superseded.

---

# 17. Validation Requirements

## 17.1 Pose Validation

Metrics:

- Pixel error.

- Normalized keypoint error.
- PCK.
- Missing-landmark rate.
- Track continuity.
- Left-right identity errors.

## 17.2 Event Validation

Metrics:

- Contact-frame mean absolute error.
- Stroke-boundary error.
- Event precision.
- Event recall.
- Event F1.
- Classification confusion matrix.

## 17.3 Court Validation

Metrics:

- Corner reprojection error.
- Court-coordinate error.
- Homography consistency.

## 17.4 Metric Validation

Metrics:

- Joint-angle error.
- Timing error.
- Repeatability.
- Intraclass correlation where applicable.
- Coefficient of variation.
- Bland–Altman analysis for reference comparisons.

## 17.5 Product Validation

Questions:

- Can the user understand why a finding appeared?
- Can the user locate its evidence?
- Can the user correct an incorrect result?
- Does the same video produce the same result under the same version?
- Does a model update clearly identify changed results?
- Are unsupported metrics withheld?

## 17.6 Dataset Split Rule

Train, validation, and test data shall be separated by athlete or session, not randomly by frame.

---

# 18. Non-Functional Requirements

## 18.1 Performance

- UI interactions shall remain responsive during browser preview.
- Heavy analysis shall not run inside Vercel Functions.
- Long-running processing shall run in the local worker or future dedicated worker.
- Browser pose inference shall use a Web Worker.
- Analysis progress shall be visible by pipeline step.
- Cancellation shall be supported where technically safe.

A concrete runtime target shall be established after benchmarking on the developer's reference computer.

## 18.2 Reliability

- Jobs shall be retryable.
- Artifacts shall use content hashes.
- Partial outputs shall be identifiable.
- Failed steps shall not mark the full run as successful.
- The user shall see actionable error information.
- Imported packages shall be schema validated.

## 18.3 Security

- Private by default.
- Signed artifact access.
- Row-level authorization.
- No public storage bucket for personal video.
- No service-role key inside browser code.
- Worker authentication shall use a scoped token.
- Secrets shall remain outside the repository.
- Logs shall not contain signed URLs or raw credentials.

## 18.4 Privacy

- Original video remains local by default.
- Face blur shall be added as a future optional processing step.
- Training-data use requires separate opt-in.
- User may delete sessions and cloud metadata.
- Multi-person videos shall carry a consent warning.

- Exported reports shall indicate whether faces are visible.

## 18.5 Reproducibility

Same input, same configuration, same model, and same pipeline version shall produce equivalent deterministic metrics within documented numeric tolerances.

## 18.6 Accessibility

- Keyboard-operable video controls.
- Text alternative for visual findings.
- Confidence is represented by labels, not color alone.
- Charts provide values in text or table form.
- Mobile layout supports result review, although detailed annotation may prioritize desktop.

## 18.7 Compatibility

Initial target:

- Current Chromium-based desktop browser.
- Windows local worker.
- Responsive mobile review.

Future targets:

- macOS worker.
- Linux worker.
- Android capture assistant.
- iOS capture assistant.

---

# 19. Cost-Constrained Architecture

## 19.1 Mandatory Cost Target

| Item              | Initial target |
|-------------------|----------------|
| Web hosting       | Rp0/month      |
| Authentication    | Rp0/month      |
| Database          | Rp0/month      |
| Raw video storage | Rp0/month      |
| Analysis compute  | Rp0/month      |

| Item                                 | Initial target   |
|--------------------------------------|------------------|
| LLM usage                            | Rp0/month        |
| Cloud GPU                            | Rp0/month        |
| Custom domain                        | Not required     |
| Total mandatory recurring cloud cost | <b>Rp0/month</b> |

Electricity, internet access, and depreciation of the user's existing computer are not included in the cloud budget.

## 19.2 Web Hosting

Use Vercel Hobby for the personal, non-commercial version. Vercel describes Hobby as a free plan for personal projects and small-scale applications, with non-commercial personal-use restrictions.

### Constraints

- Do not run video inference in Vercel Functions.
- Do not depend on long-running server functions.
- Do not treat Hobby as the future commercial hosting plan.
- Upgrade or migrate only when the product becomes commercial.

## 19.3 Database and Authentication

Use one Supabase Free project for:

- Authentication.
- PostgreSQL metadata.
- Analysis records.
- Metrics.
- Annotations.
- Small thumbnails or exports.

As of this PRD, Supabase Free includes 500 MB database storage, 1 GB file storage, 5 GB egress, and 50,000 monthly active users. Free projects may pause after one week of inactivity, and free Storage has a 50 MB maximum file-upload size.

### Implication

Raw badminton videos shall not depend on Supabase Storage. The free tier is sufficient for structured data and limited small artifacts, not a growing video library.

## 19.4 Raw Video Storage

### Default

- Video stays on the local computer.
- Worker accesses local files.
- Cloud stores file fingerprint and metadata.
- User remains responsible for local backup.

### Optional Cloud Storage

Cloudflare R2 may be added for selected proxy videos or manually synced sessions. Its current Standard free tier includes 10 GB-month storage, one million Class A operations, ten million Class B operations, and free internet egress.

R2 is optional and shall not be required for the initial release.

## 19.5 Compute

### Initial

- Python local worker.
- CPU support required.
- NVIDIA GPU acceleration optional.
- Model quality may be selectable based on available hardware.

### Future

- Pay-per-job GPU.
- Self-hosted workstation.
- Dedicated cloud worker.
- Community research compute.

No cloud GPU shall remain running while idle.

## 19.6 LLM

Initial implementation uses deterministic template-based explanations.

An LLM may later be enabled with:

- User-provided API key.
- Local model.
- Limited manual invocation.
- Explicit cost indication.

Core metric calculation shall never depend on an LLM.

## 19.7 Cost Guardrails

- No cloud resource with automatic unlimited billing.
  - No persistent GPU instance.
  - No raw-video auto-upload.
  - No paid observability dependency.
  - No paid image/video transformation.
  - No daily scheduled model processing.
  - No automatic re-analysis of all historical videos.
  - Optional cloud features must be disabled by default.
  - Storage retention must be configurable.
  - Derived frames should be deleted after packaging unless retained explicitly.
- 

## 20. Recommended Technical Stack

### 20.1 Web Application

- Next.js.
- TypeScript.
- Tailwind CSS.
- shadcn/ui.
- Supabase client.
- Canvas or WebGL overlay renderer.
- TanStack Query.
- Zod schemas.

### 20.2 Local Worker

- Python.
- FastAPI local service or CLI.
- FFmpeg.
- OpenCV.
- NumPy.
- SciPy.
- PyTorch.
- MMPose or compatible pose adapter.
- MediaPipe as lightweight fallback.
- Polars or DuckDB for result processing.

### 20.3 Shared Contracts

- JSON Schema or Zod-generated contracts.
- Versioned analysis manifest.
- Versioned artifact manifest.



- Versioned metric catalogue.

## 20.4 Repository

Recommended monorepo:

```
badminton-motion-lab/  
├─ apps/  
│   ├── web/  
│   └─ research-admin/  
├─ workers/  
│   └─ local-analyzer/  
├─ packages/  
│   ├── domain/  
│   ├── analysis-contracts/  
│   ├── badminton-ontology/  
│   ├── metric-engine/  
│   ├── coordinate-systems/  
│   ├── database/  
│   └─ ui/  
├─ ml/  
│   ├── adapters/  
│   ├── models/  
│   ├── datasets/  
│   ├── evaluation/  
│   └─ experiments/  
├─ validation/  
├─ docs/  
└─ infra/
```

---

## 21. Release Phases

### Phase 0 — Product and Scientific Foundation

#### Objective

Establish stable product language and data contracts.

#### Deliverables

- Finalized PRD.
- Architecture specification.

- Badminton ontology.
- Metric catalogue.
- Coordinate-system specification.
- Analysis contract.
- Data model and ERD.
- Validation plan.
- Cost guardrails.
- Repository skeleton.
- Architecture decision records.

#### **Exit criteria**

- One representative stroke can be mapped from video to observation, metric, finding, and evidence.
- All major entities have clear ownership and versioning.
- No raw-video cloud dependency exists.

---

## **Phase 1 — Local Analysis Backbone**

### **Objective**

Create a complete but model-light end-to-end flow.

### **Deliverables**

- Session CRUD.
- Athlete profile.
- Local file selection.
- Media metadata extraction.
- Analysis job creation.
- Local Python worker.
- Analysis package import.
- Analysis-run versioning.
- Basic video viewer.
- Frame stepping.
- Manual contact and stroke markers.
- JSON and CSV export.

### **Initial analytical capability**

- No automatic technical judgment required.
- Manual event annotation is acceptable.
- Mock or baseline pose results are acceptable.

### **Exit criteria**

- A local video can produce a versioned analysis package.

- The web application can display the result.
  - Reprocessing creates a new run.
  - Original result remains accessible.
  - Monthly cloud cost remains Rp0.
- 

## Phase 2 — Pose, Calibration, and Basic Metrics

### Objective

Deliver the first genuinely useful personal analysis.

### Deliverables

- MediaPipe preview.
- Local detailed pose adapter.
- Manual court calibration.
- Pose smoothing.
- Joint-angle metrics.
- Body-center trajectory.
- Basic footwork path.
- Quality flags.
- Evidence-linked metrics.
- Basic session comparison.
- Manual annotation correction.

### Exit criteria

- Joint metrics are reproducible.
  - Court-plane path is displayed.
  - Unsupported metrics are withheld.
  - Same video produces equivalent metrics under the same pipeline version.
- 

## Phase 3 — Automated Event Intelligence

### Objective

Reduce manual annotation.

### Deliverables

- Stroke proposal.
- Contact proposal.
- Movement-phase detection.
- Audio impact features.

- Repetition grouping.
- Basic stroke taxonomy.
- Confidence-aware correction workflow.

#### **Exit criteria**

- Automatic proposals reduce manual review effort.
  - Contact-frame error is measured on an internal benchmark.
  - User correction remains fast and non-destructive.
- 

## **Phase 4 — Racket and Shuttle Intelligence**

### **Objective**

Connect body movement to racket and shuttle events.

### **Deliverables**

- Manual racket annotation.
- Racket detector.
- Racket trajectory.
- Shuttle tracker.
- Track rectification.
- Contact confidence.
- Racket-body timing metrics.
- Better stroke classification.

### **Exit criteria**

- Racket and shuttle benchmarks are documented.
  - Missing and interpolated tracks are clearly labeled.
  - Contact estimation improves relative to Phase 3.
- 

## **Phase 5 — Technique and Footwork Lab**

### **Objective**

Provide structured training analysis.

### **Deliverables**

- Full metric catalogue for selected strokes.
- Footwork event taxonomy.
- Recovery analysis.

- Repetition consistency.
- Personal baseline.
- Evidence-backed findings.
- Structured training priorities.
- Session report.

#### **Exit criteria**

- Each finding has metric, evidence, reference, confidence, and limitation.
  - User can compare equivalent sessions.
  - Repeatability meets documented thresholds.
- 

## **Phase 6 — Multi-Camera Precision**

### **Objective**

Improve 3D measurement validity.

### **Deliverables**

- Multi-video session.
- Synchronization.
- Camera calibration.
- Triangulation.
- Anatomical constraints.
- 3D pose.
- 3D racket orientation.
- 3D contact position.
- Validation against reference measurements.

### **Exit criteria**

- Multi-camera metric errors meet predefined validation targets.
  - Single-camera and multi-camera outputs are clearly distinguished.
- 

## **Phase 7 — Match Intelligence**

### **Objective**

Analyze rallies and full matches.

### **Deliverables**

- Match/game/rally hierarchy.

- Multi-player tracking.
- Rally segmentation.
- Stroke sequence.
- Court-zone mapping.
- Shot outcome.
- Tactical tendencies.
- Match report.
- Singles support.
- Doubles support later within the same domain model.

#### **Exit criteria**

- A match can be represented as a corrected structured dataset.
- Tactical insights link to strokes and video evidence.

---

## **Phase 8 — Coach and Personalization**

### **Objective**

Add structured expert review and personalized assessment.

### **Deliverables**

- Coach relationship.
- Coach comments.
- Custom rubrics.
- Reference movement.
- Coach-approved findings.
- Athlete-specific thresholds.
- Training targets.
- Progress alerts.
- Grounded LLM explanation.

### **Exit criteria**

- Measurement and coach judgment remain distinct.
- Personal thresholds are based on sufficient valid observations.

---

## **Phase 9 — Guided Capture and Near-Real-Time Analysis**

### **Objective**

Improve input quality before processing.

### **Deliverables**

- Camera placement guide.
- Live body framing.
- Court visibility check.
- Blur and lighting warning.
- Lightweight pose preview.
- Automated capture profile.
- Near-real-time drill feedback.

### **Exit criteria**

- New users can produce analyzable recordings without technical assistance.
  - Capture failures decrease against the previous baseline.
- 

## **Phase 10 — Research-Grade Validation**

### **Objective**

Support credible claims of accuracy.

### **Deliverables**

- Gold dataset.
- Expert annotation protocol.
- Multiple-reviewer agreement study.
- Device robustness study.
- Camera-angle study.
- Multi-camera validation.
- Model cards.
- Metric cards.
- Reproducible benchmark runner.
- Badminton-specific fine-tuning.

### **Exit criteria**

- Every published accuracy claim includes dataset scope, method, and limitation.
  - Product confidence is calibrated against observed error.
- 

## **22. Initial Release Definition**

The first release is not considered the final MVP boundary. It is the first usable implementation of the complete architecture.

## Included

- Authentication.
- Athlete profile.
- Session management.
- Local video analysis.
- Media metadata.
- Local analysis worker.
- Baseline pose estimation.
- Manual court calibration.
- Manual stroke range.
- Manual contact frame.
- Skeleton overlay.
- Basic body angles.
- Body path.
- Quality flags.
- Analysis versioning.
- JSON/CSV export.
- Annotation revision.
- Session history.

## Not Yet Active but Architecturally Represented

- Racket tracks.
- Shuttle tracks.
- Match/game/rally.
- Multi-camera groups.
- Coach review.
- Assessment rubrics.
- Model registry.
- Validation datasets.
- LLM explanations.
- Cloud workers.

Database entities and contracts for these areas may exist before their UI is enabled.

---

## 23. Initial Success Metrics

### 23.1 Product Usability

- A new session can be created without database intervention.
- A video can be analyzed locally.
- Results can be reviewed in the browser.
- User can identify the source frame for each metric.
- User can correct contact frame.



- User can export a complete analysis package.

## 23.2 Technical Reliability

- At least 95% of supported test clips complete the baseline pipeline.
- Re-running the same version does not create inconsistent deterministic metrics.
- Failed jobs expose the failing step.
- No raw video is unintentionally uploaded.
- No required paid service is invoked.

## 23.3 Cost

- Mandatory monthly cloud expenditure remains Rp0.
- No cloud GPU usage.
- Database remains within free-tier limits.
- No raw-video dependency on Supabase Storage.
- No LLM API cost.

## 23.4 Analytical Integrity

- Every displayed metric has a definition.
  - Every metric identifies its source frames.
  - Every low-confidence result is labeled.
  - Unsupported analysis is not presented as successful.
  - Model output and human correction remain separate.
- 

# 24. Risks and Mitigations

## Risk 1 — Local Worker Is Less Convenient

### Mitigation

- Provide one-command installation.
- Add clear connection status.
- Support portable analysis packages.
- Later provide an optional cloud worker.

## Risk 2 — Free Database Project Pauses

### Mitigation

- Keep local analysis functional.
- Use schema migrations and reproducible seed data.
- Treat cloud sync as a convenience rather than the only data source.

- Export analysis packages for backup.

### **Risk 3 — Video Storage Grows Quickly**

#### **Mitigation**

- Keep originals local.
- Store only metadata and metrics in cloud.
- Use lifecycle deletion for derived frames.
- Make cloud media synchronization opt-in.

### **Risk 4 — CPU Processing Is Slow**

#### **Mitigation**

- Provide fast, standard, and detailed profiles.
- Process cropped player regions.
- Cache intermediate artifacts.
- Allow GPU acceleration when available.
- Run expensive modules only when requested.

### **Risk 5 — Baseline Pose Is Inaccurate for Badminton**

#### **Mitigation**

- Allow correction.
- Benchmark multiple models.
- Collect difficult-frame dataset.
- Use model adapter.
- Fine-tune only after enough annotated data exists.

### **Risk 6 — User Mistakes Confidence for Accuracy**

#### **Mitigation**

- Separate model confidence from validated error.
- Explain confidence meaning.
- Publish metric limitations.
- Do not display unnecessary decimal precision.

### **Risk 7 — Scope Becomes Too Large**

#### **Mitigation**

- Maintain the complete domain model.
- Activate capabilities by phase.

- Use exit gates.
- Avoid building UI for inactive modules.
- Prioritize end-to-end vertical slices.

## **Risk 8 — LLM Introduces Unreliable Feedback**

### **Mitigation**

- Keep LLM optional.
- Use deterministic explanation templates first.
- Ground all generated text in structured findings.
- Validate every numeric reference.

## **Risk 9 — Project Becomes Expensive After Cloud Migration**

### **Mitigation**

- Preserve local execution as a permanent mode.
- Use cloud compute only per job.
- Add per-job cost estimation.
- Require explicit opt-in before paid processing.
- Use storage abstraction and worker adapter.

---

# **25. Open Product Decisions**

The following decisions should be resolved during technical design rather than blocking the PRD:

1. Whether the local worker begins as CLI, desktop tray app, or localhost service.
  2. Whether local analysis packages are ZIP archives or directory manifests.
  3. Which baseline pose checkpoint provides the best accuracy-to-runtime ratio.
  4. Whether the initial viewer uses Canvas 2D or WebGL.
  5. Whether local file references use browser APIs or worker-controlled file selection.
  6. Which subset of technique metrics is valid for each camera angle.
  7. Whether SQLite is used as an additional local catalogue.
  8. Whether cloud metadata sync is automatic or manual.
  9. Which internal annotation format becomes canonical.
  10. Whether the future desktop wrapper uses Tauri or remains browser-plus-worker.
-

## 26. Definition of Done for PRD

This PRD is considered approved when:

- Product vision is agreed.
- Rp0 operating constraint is accepted.
- Local-first strategy is accepted.
- Initial release scope is agreed.
- Product phases are agreed.
- Non-goals are accepted.
- Core functional requirements are complete enough to create an ERD.
- Metric and validation details can be moved into separate specifications.
- No major product capability requires restructuring the proposed domain model.